

Module 4

Shell Productivity & Basic Scripting

Learning objectives

Use history and Ctrl+R effectively

Write scripts with shebang and chmod +x

Pass arguments with \$1 and \$@

Apply set -e and quoted variables

Prerequisites

Module 3 (redirection, pipes)

Learning path

Learning Path

Diagram: Learning Path
(render with mmdc when available)
flowchart LR
A[History & aliases] --> B[Scripts]
B --> C[Variables]
C --> D[Control flow]
D --> E[Safe scripting]
E --> F[Module 5]

Course context

Wrap compile/sim steps in scripts/

Course module scripts are the pattern to follow

Quote variables to avoid word-splitting bugs

Core topics

Self-check

```
$ ./scripts/module4.sh --check
```

Terminal: module_check

```

[0;36m=====
[0;36mModule 4: Environment self-check
[0;36m=====

[0;32m[OK] bash available
[0;32m[OK] module4 directory exists: /mnt/d/proj/universal-verification-methodology/learn_unix_
[0;32m[OK] script_basics/say_hello.sh exists
[0;32m[OK] arguments/greet.sh exists
[0;32m[OK] control_flow/count_lines.sh exists
[0;32m[OK] safe_scripting/clean_tmp.sh exists

[0;32m[OK] All required checks passed.
```

Minimal script

```
#!/usr/bin/env bash  
echo "Hello from script"
```

First bash script

```
$ cd module4/examples/script_basics && chmod u+x say_hello.sh && ./say_hello.sh
```

```
Terminal: say_hello  
Hello from the script!
```

Positional parameters

```
$ cd module4/examples/arguments && chmod u+x greet.sh && ./greet.sh Student
```

```
Terminal: greet_args  
Hello, Student!
```

Practice

Exercise scaffold

```
./scripts/module4.sh --scaffold
```

Exercises

Run `say_hello.sh` and `greet.sh` from scaffold

Add an alias `ll='ls -aF'`

Write a loop over `*.log`

Summary & next steps

Scripts automate design flows

History and aliases save time

Next: Module 5 — projects & archives

Next: Module 5: Editors, Projects & Archives